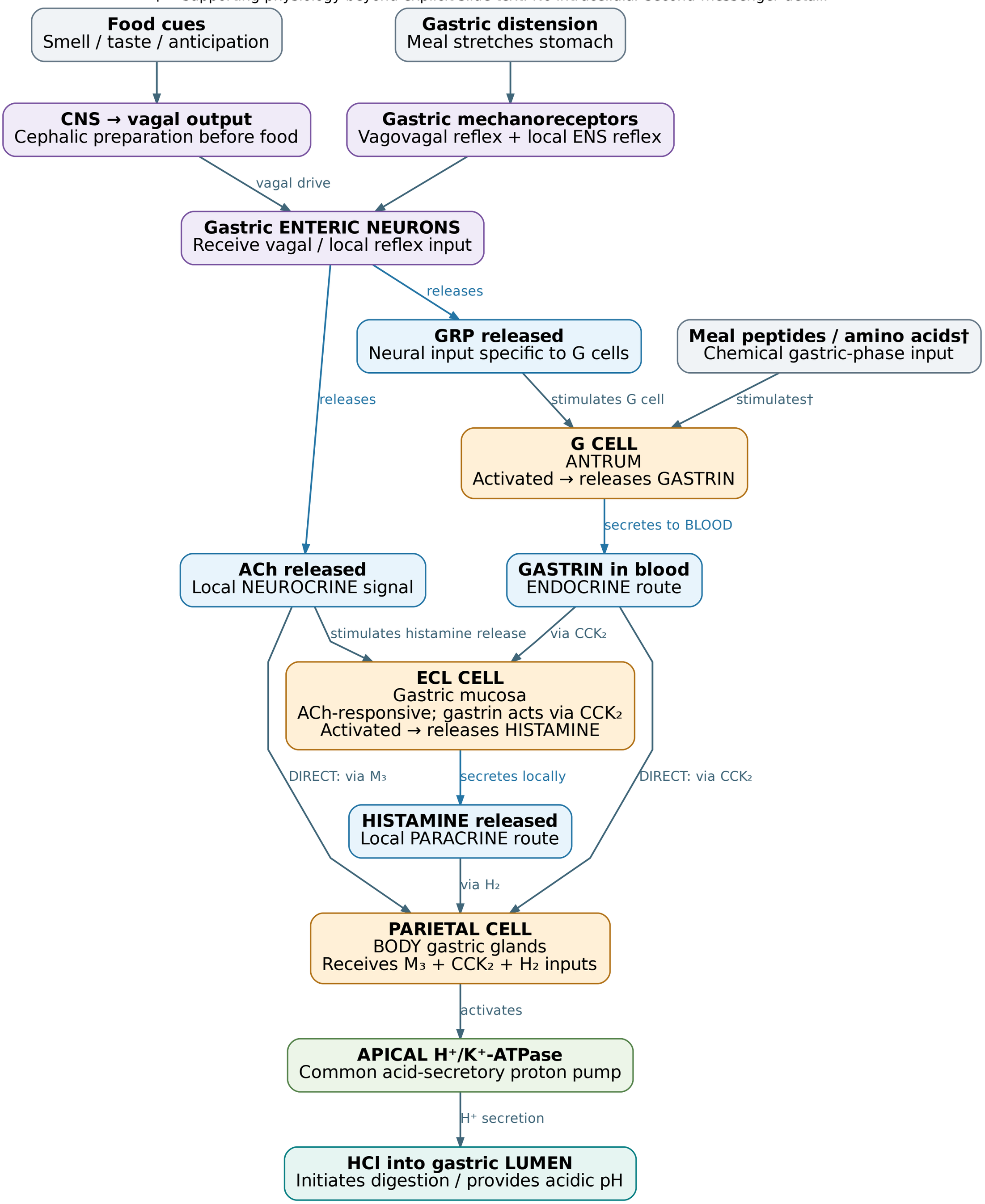
01. GASTRIC ACTIVATION: cells, secretions and direct / indirect routes

Lecture PDF pp. 11–14, 16; cephalic + gastric phases

Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.

† = supporting physiology beyond explicit slide text. No intracellular second-messenger detail.



Read the branches: ACh directly stimulates parietal cells AND ECL cells. GRP stimulates G cells; gastrin then acts directly on parietal cells AND indirectly through ECL histamine.

Cell origins: G cells—antrum; ECL and parietal cells—gastric mucosa / body glands. Molecule release is distinct from the responding receptor on another cell.

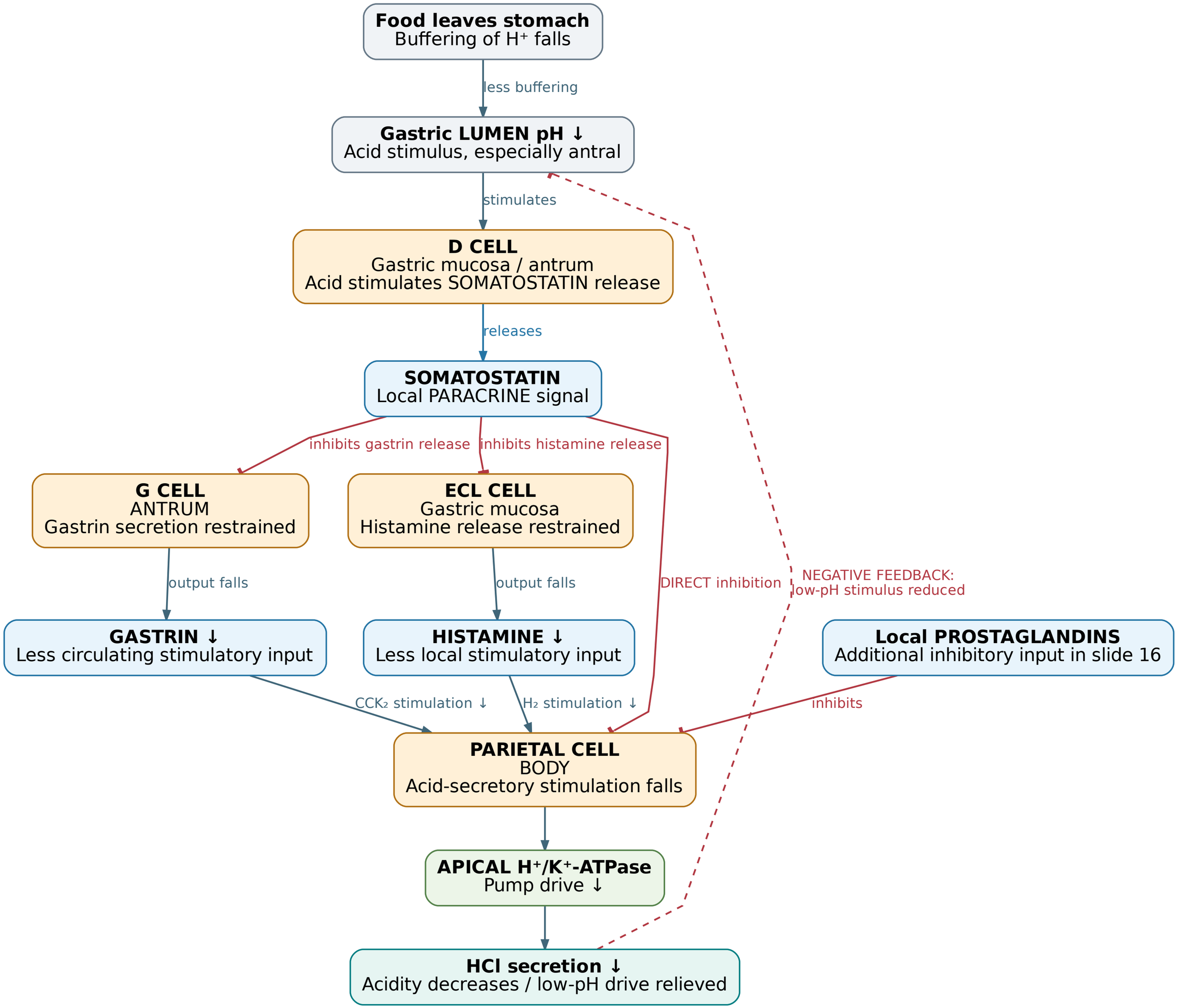
† Meal peptides / amino acids → G-cell stimulation supplies a supporting physiological link. Strong gastric acidity is an inhibitory D-cell stimulus (Figure 2), not the gastric-phase stimulus printed on slide 12.

02. GASTRIC BRAKE: D-cell somatostatin and negative feedback

Lecture PDF pp. 15–16; correction to the low-pH wording on p. 12

Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.

† = supporting physiology beyond explicit slide text. No intracellular second-messenger detail.



DIRECT inhibition: somatostatin → parietal cell. INDIRECT inhibition: somatostatin → G cell / ECL cell → less gastrin / histamine → less parietal stimulation.

The red feedback return means that reduced acid output relieves the low-pH stimulus. It does not mean somatostatin causes additional acidification.

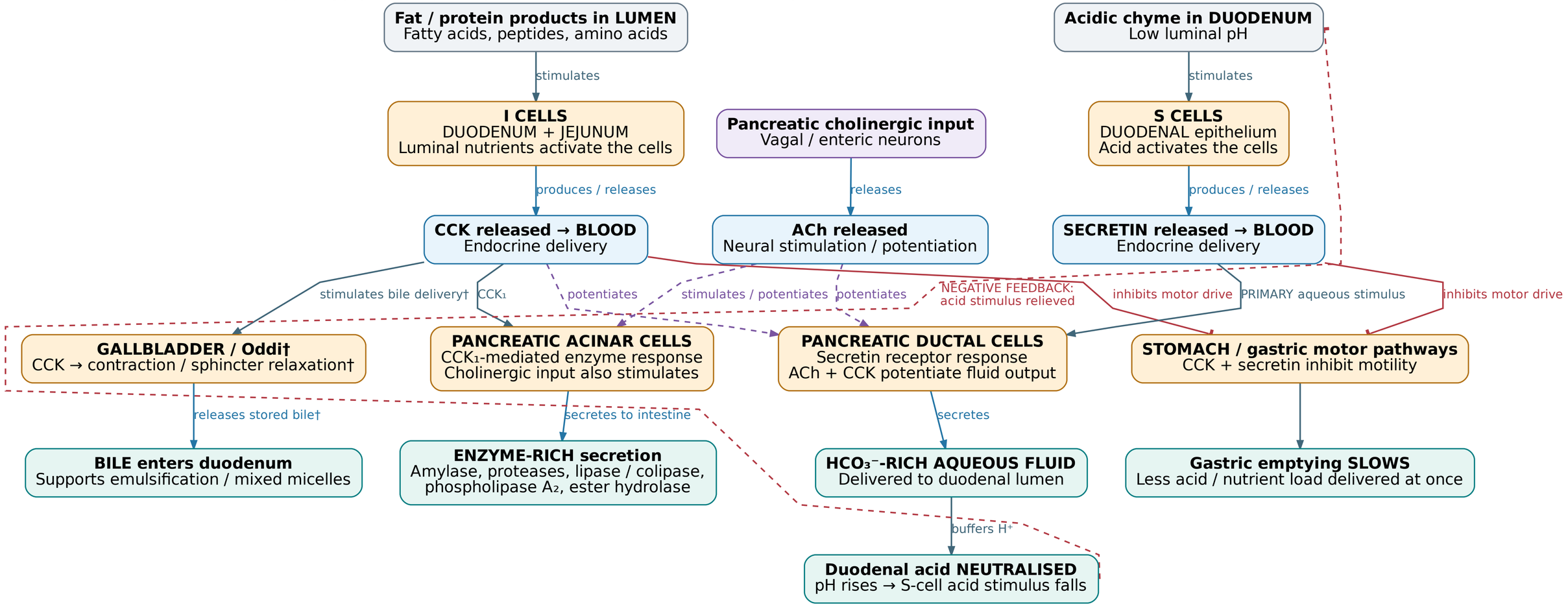
Food initially buffers H^+. As food leaves, buffering falls and gastric pH can fall more, strengthening the D-cell brake. Basal and meal-related secretion are regulated, not simply switched off.

03. INTESTINE–PANCREAS: sensing cells, hormone outputs and organ interactions

Lecture PDF pp. 17–19, 23; parallel endocrine and neural inputs

Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.

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I cells respond to luminal amino acids, peptides and fatty acids; S cells respond to DUODENAL ACID. Their released hormones enter blood and reach distinct effector cells.

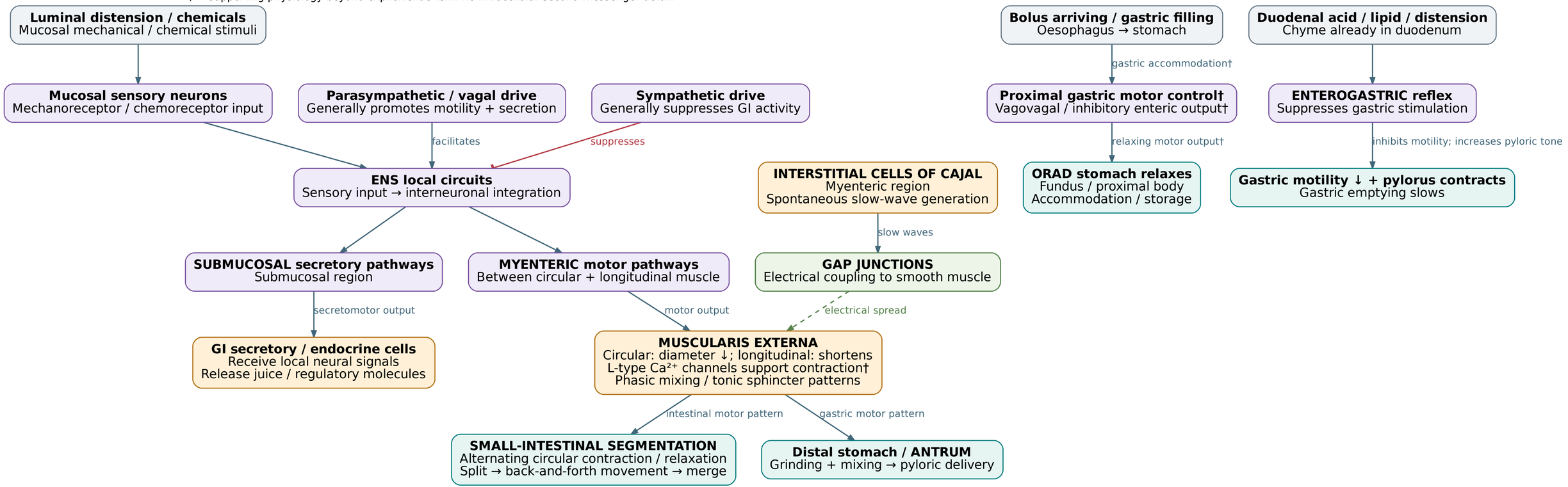
CCK_1–acinar secretion follows the introductory lecture model. ACh and CCK support pancreatic secretion together; their cooperation is not by itself positive feedback.

† CCK-driven gallbladder contraction / Oddi relaxation completes the bile-delivery link, which is not developed explicitly in this lecture. Bile is made in LIVER and stored in GALLBLADDER.

04. NEURAL & MOTOR NETWORK: sensory input, ENS and smooth-muscle effects

Lecture PDF pp. 4–10, 12, 17

Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.
† = supporting physiology beyond explicit slide text. No intracellular second-messenger detail.



Myenteric plexus lies BETWEEN circular and longitudinal muscle; submucosal plexus lies internal to circular muscle in the submucosal region. ENS circuits can operate locally.

GASTRIC distension promotes gastric-phase activity; DUODENAL distension / acid / lipid creates an upstream enterogastric brake. These are different anatomical stimuli.

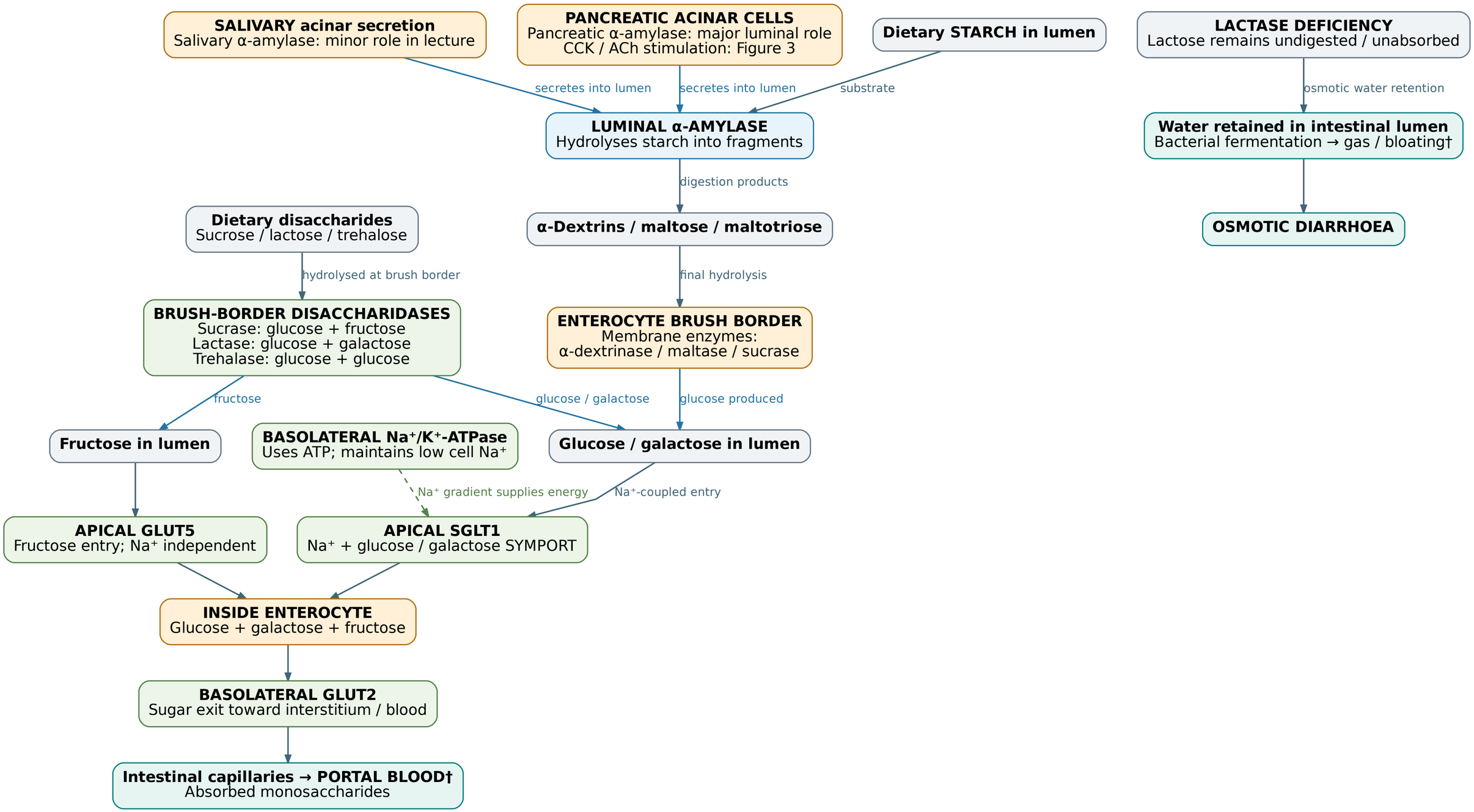
ICCs are pacemaker cells, not enteric neurons. Slow waves coordinate excitability; sufficient depolarisation permits strong contraction. † identifies a supporting inhibitory-motor link for accommodation.

05. CARBOHYDRATE: enzyme sources → brush border → enterocyte → portal blood

Lecture PDF pp. 20–22; digestion, absorption and lactase deficiency

Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.

† = supporting physiology beyond explicit slide text. No intracellular second-messenger detail.



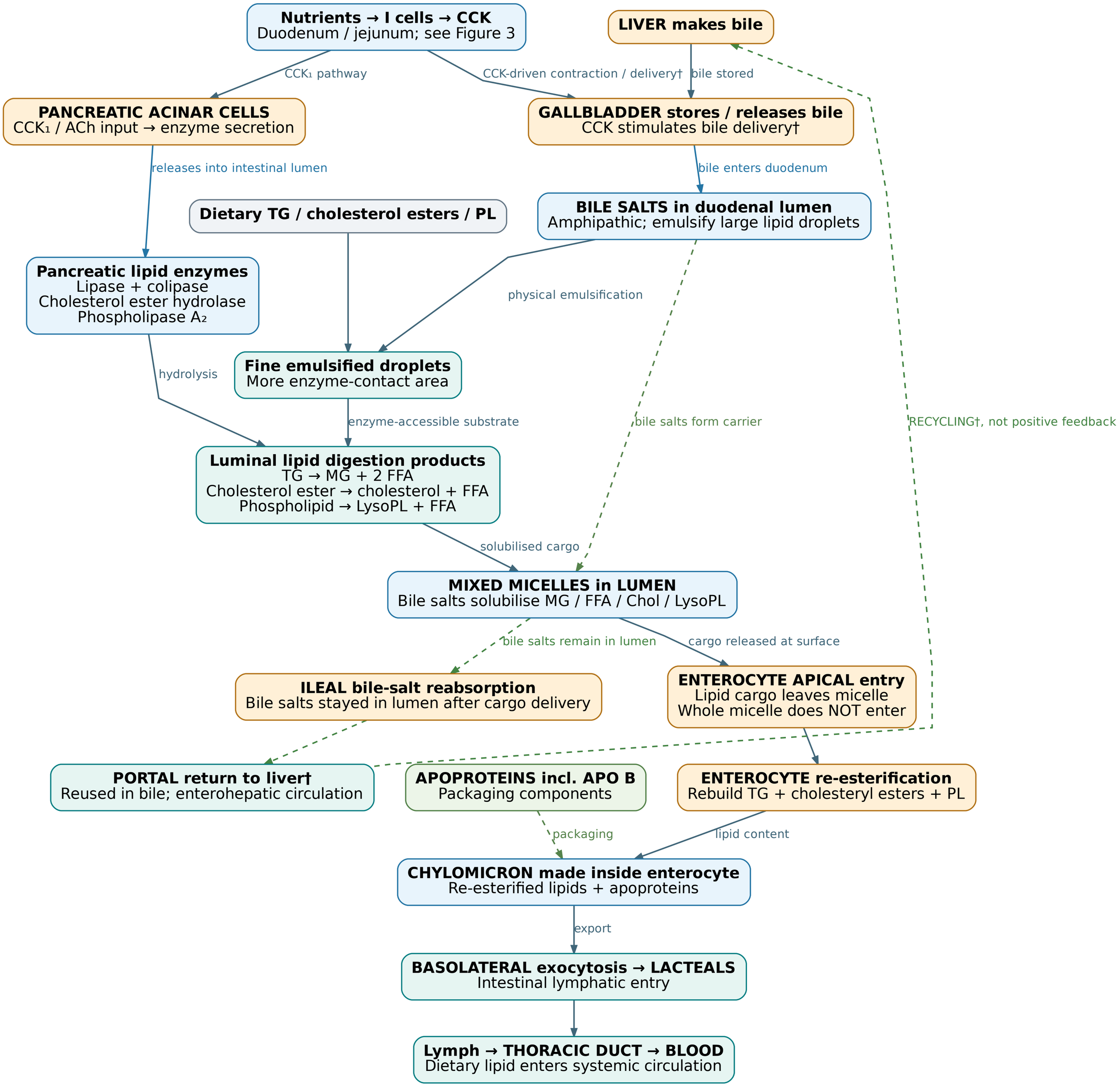
Correction retained: salivary / pancreatic α-amylase acts in the LUMEN. Enterocyte BRUSH-BORDER enzymes, not a major mucosal α-amylase secretion, complete digestion.

The Na^+/K^+-ATPase supplies the Na^+ gradient; SGLT1 couples sugar entry to Na^+ movement. GLUT5 and GLUT2 provide facilitated diffusion. This is energy coupling, not positive feedback.

Only monosaccharides enter the absorption pathway shown here. The source maltotriose arrow names sucrase; brush-border α-glucosidases have overlapping activity. † Portal delivery / fermentation are supporting links.

06. LIPID: bile / enzyme sources, enterocyte processing and lymphatic transport

Lecture PDF pp. 20, 23–25; repeated CCK link shown explicitly
Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.
† = supporting physiology beyond explicit slide text. No intracellular second-messenger detail.



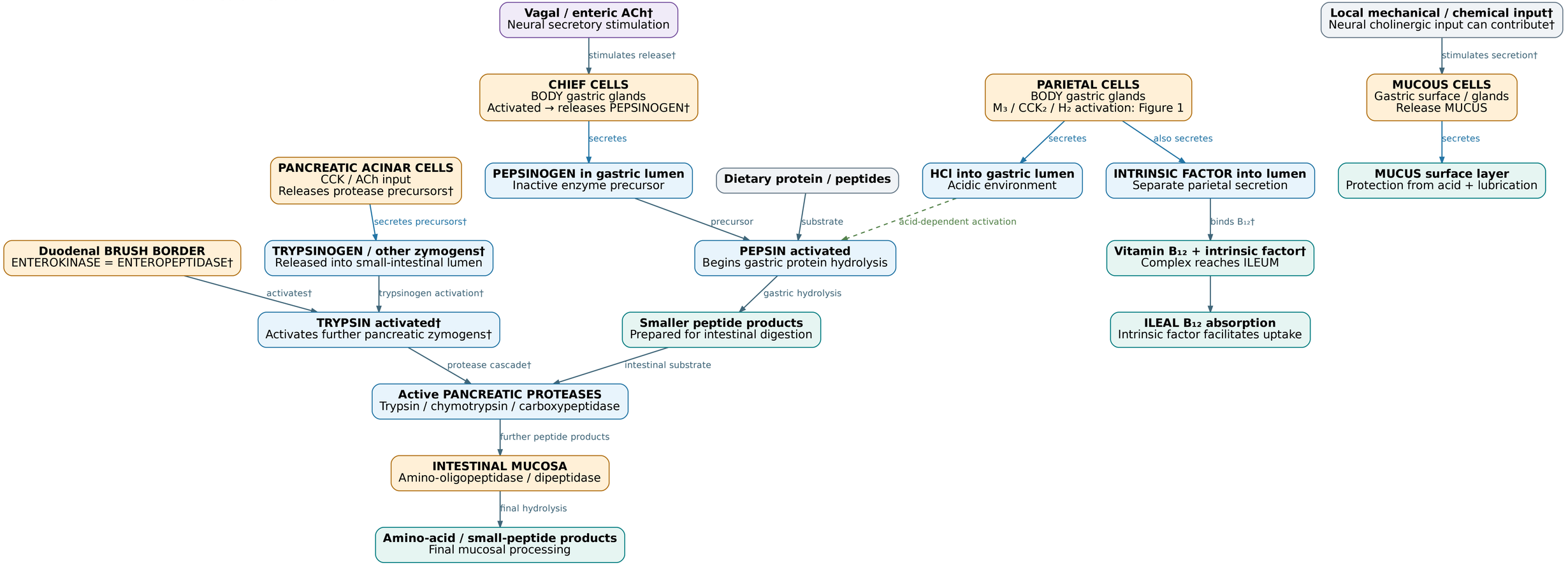
Bile salts EMULSIFY; pancreatic enzymes HYDROLYSE. Micelle cargo enters the cell; the whole micelle / bile salts remain outside. The source also lists oral LINGUAL LIPASE for preliminary lipid digestion.

Inside the enterocyte: monoglycerides + acyl chains → triglycerides; cholesterol → cholesteryl esters; lysophospholipids → phospholipids. Apo B supports chylomicron packaging.

Green dashed return is bile-salt RECYCLING, not positive feedback. † CCK-driven bile delivery and portal return complete supporting physiological links. The lymph route mainly describes long-chain dietary lipid.

07. OTHER GASTRIC CELLS & PROTEIN ENZYMES: stimulus → product → action

Lecture PDF pp. 11, 20; † fills supporting activation links
Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.
† = supporting physiology beyond explicit slide text. No intracellular second-messenger detail.



The lecture explicitly gives the cell products and enzyme list. † supplies supporting chief-cell / mucous-cell stimulation and protease-activation links; it is not labelled as extra slide content.

HCl is supplied by parietal cells, pepsinogen by chief cells: acid activates the precursor and supports pepsin action. Intrinsic factor is a separate parietal product, not the output of the H^+/K^+ pump.

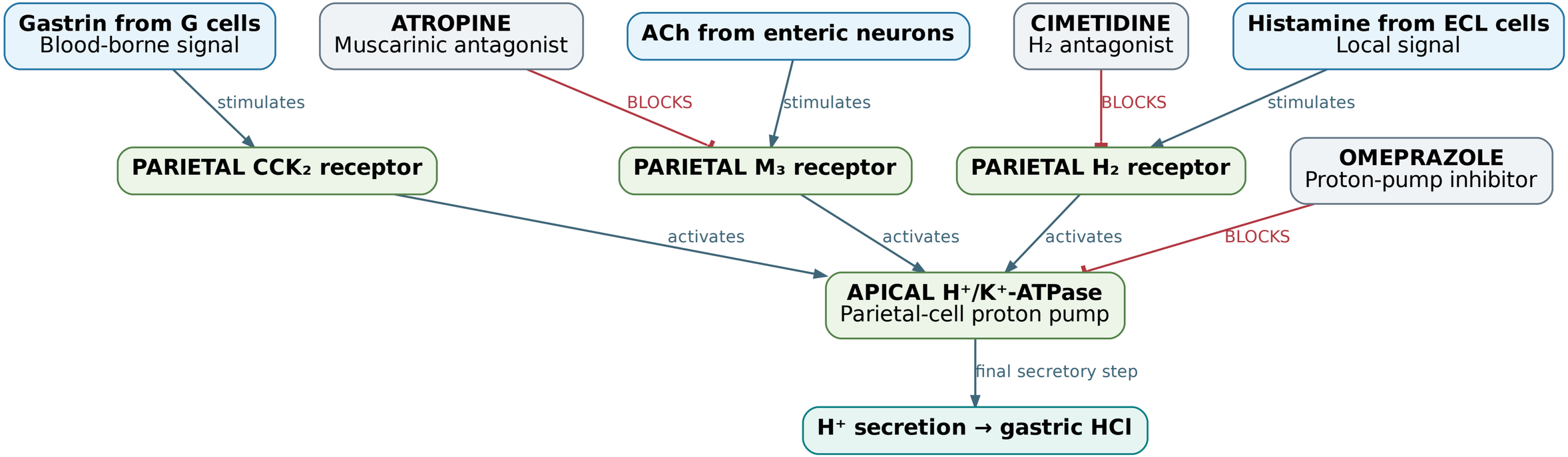
The lecture focuses detailed absorption mechanisms on carbohydrates and lipids. No additional amino-acid / peptide transporter syllabus is assumed here.

08. RECEPTOR & PUMP BLOCKADE: where each named drug interrupts the network

Lecture PDF p. 16; three distinct pharmacological targets

Cells are amber; released molecules are blue; key targets / proteins are green. Red flat ends inhibit.

† = supporting physiology beyond explicit slide text. No intracellular second-messenger detail.



Atropine blocks muscarinic signalling, including parietal M_3. Cimetidine blocks H_2. Omeprazole inhibits the final H^+/K^+-ATPase, downstream of the converging receptor inputs.

Receptor blockade leaves the other receptor systems present; pump inhibition can lower acid output despite continued upstream stimuli. These drug actions are not themselves feedback loops.

For the regulatory loop, return to Figure 2: acidity activates D cells, which release somatostatin to inhibit G cells, ECL cells and parietal cells.